



ITEC632 - Applied Data Mining

Semester 1 - 2025

Assessment 3

Predictive Data Mining Project

Submitted By

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Student ID

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Word Count: 1778

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1. Introduction

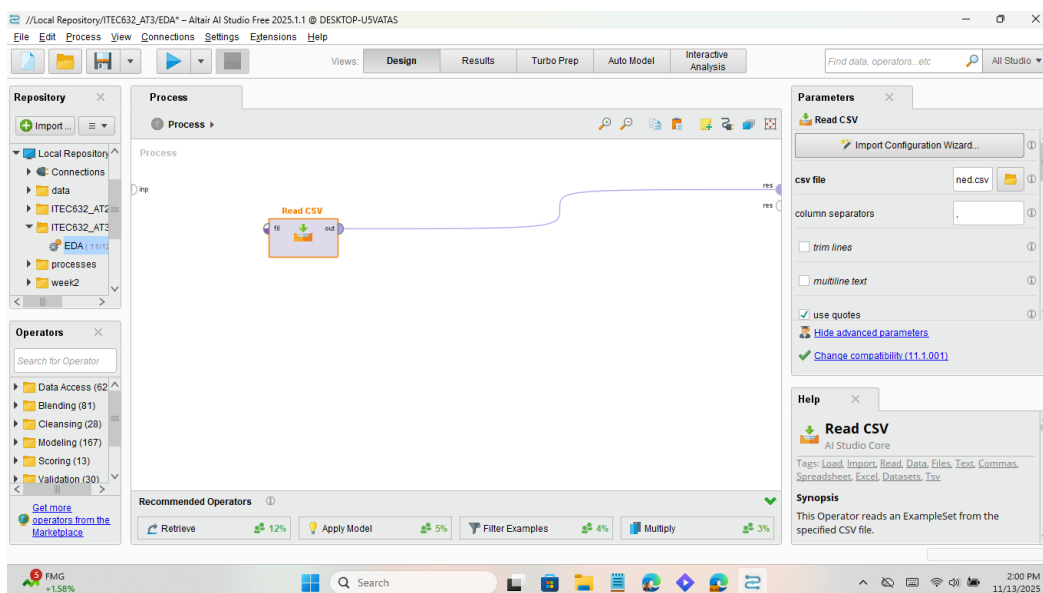
Heart disease remains one of the leading causes of morbidity and mortality worldwide, making early detection and risk assessment a critical public health priority. Advances in machine learning provide opportunities to develop predictive models that can identify individuals at risk based on lifestyle and clinical attributes. This project applies data mining techniques using RapidMiner to evaluate two models, Decision Tree and Logistic Regression, for heart disease prediction. The workflow includes preprocessing categorical variables, applying cross-validation to ensure robust evaluation, and extracting performance metrics such as accuracy, precision, recall, F-measure, AUC, and Kappa. By comparing model outputs and interpreting variable importance, the project aims to determine which approach offers the best balance between predictive performance and interpretability. Beyond technical evaluation, ethical considerations such as fairness, privacy, transparency, accountability, and potential harms are explored to ensure responsible deployment of AI in healthcare. Together, these steps provide a comprehensive framework for assessing both the technical and ethical suitability of machine learning models in clinical screening contexts.

2. Exploratory Data Analysis

a. Load and Inspect the dataset

The original huge dataset could be loaded in RapidMiner, but due to my license being a standard one, I could not process the whole set of data, so I trimmed the data down to 9500 rows.

a. Read CSV



Result:

Altair AI Studio Free 2025.1.1 @ DESKTOP-USVATAS

File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

Find data, operators, etc. All Studio

Result History ExampleSet (Read CSV)

Open in Turbo Prep Auto Model Interactive Analysis Filter (9,500 / 9,500 examples): all

Row No.	HeartDisease	BMI	Smoking	AlcoholDrim...	Stroke	PhysicalHea...	MentalHealth	DiffWalking	Sex
1	No	16.500	Yes	No	No	3	30	No	Fem
2	No	20.340	No	No	Yes	0	0	No	Fem
3	No	26.580	Yes	No	No	20	30	No	Male
4	No	24.210	No	No	No	0	0	No	Fem
5	No	23.710	No	No	No	28	0	Yes	Fem
6	Yes	28.870	Yes	No	No	6	0	Yes	Fem
7	No	21.630	No	No	No	15	0	No	Fem
8	No	31.640	Yes	No	No	5	0	Yes	Fem
9	No	26.450	No	No	No	0	0	No	Fem
10	No	40.690	No	No	No	0	0	Yes	Male
11	Yes	34.300	Yes	No	No	30	0	Yes	Male
12	No	28.710	Yes	No	No	0	0	No	Fem
13	No	28.370	Yes	No	No	0	0	Yes	Male
14	No	28.150	No	No	No	7	0	Yes	Fem

ExampleSet (9,500 examples, 0 special attributes, 18 regular attributes)

Repository

- Import Data
- Training Resources (connected)
- Samples
- Community Samples (connected)
- Local Repository (Local)
- Connections
- data
- ITEC632_AT2_WhiteWine
- ITEC632_AT3
 - EDA (11/12/25 11:27 PM ~ 3 kb)
- processes
- week2
- week4
- week6
- week8
- week10
- week12
- DB (Legacy)

XAO index -0.94%

2:03 PM 11/13/2025

b. Set Role

Altair AI Studio Free 2025.1.1 @ DESKTOP-USVATAS

File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

set role X All Studio

Repository

- Import...
- Local Repository
 - Connections
 - data
 - ITEC632_AT2
 - ITEC632_AT3
 - EDA (11/12/25 11:27 PM ~ 3 kb)
 - processes
 - week2

Operators

Search for Operator

- Data Access (62)
- Blending (81)
- Cleansing (28)
- Modeling (167)
- Scoring (13)
- Validation (30)

Get more operators from the Marketplace

Process

Process

Read CSV

Set Role

Edit Parameter List: set roles

Edit Parameter List: set roles

This parameter is used to set the roles of one or more Attributes. A click on "Edit List (0)..." opens a menu with Attribute name and target role pairs. The role of an Attribute will be changed to the given target role for each of the pairs. Following target roles are possible:

attribute name	target role
HeartDisease	label

Recommended Operators

- Retrieve 75%
- Select Attributes 52%
- Apply

Temp to drop Tomorrow

2:06 PM 11/13/2025

Result:

Altair AI Studio Free 2025.1.1 @ DESKTOP-USVATAS

File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

set role X All Studio

Result History ExampleSet (Set Role)

Open in Turbo Prep Auto Model Interactive Analysis

Filter (9,500 / 9,500 examples): all

Row No.	HeartDisease	BMI	Smoking	AlcoholDrin...	Stroke	PhysicalHea...	MentalHealth	DiffWalking	Sex
1	No	16.600	Yes	No	No	3	30	No	Fem
2	No	20.340	No	No	Yes	0	0	No	Fem
3	No	26.580	Yes	No	No	20	30	No	Male
4	No	24.210	No	No	No	0	0	No	Fem
5	No	23.710	No	No	No	28	0	Yes	Fem
6	Yes	28.870	Yes	No	No	6	0	Yes	Fem
7	No	21.630	No	No	No	15	0	No	Fem
8	No	31.640	Yes	No	No	5	0	Yes	Fem
9	No	26.450	No	No	No	0	0	No	Fem
10	No	40.690	No	No	No	0	0	Yes	Male
11	Yes	34.300	Yes	No	No	30	0	Yes	Male
12	No	28.710	Yes	No	No	0	0	No	Fem
13	No	28.370	Yes	No	No	0	0	Yes	Male
14	No	28.150	No	No	No	7	0	Yes	Fem

ExampleSet (9,500 examples, 1 special attribute, 17 regular attributes)

Repository

Import Data

- Training Resources (connected)
- Samples
- Community Samples (connected)
- Local Repository (Local)
 - Connections
 - data
 - ITEC632_AT2_WhiteWine
 - ITEC632_AT3
 - EDA (11/12/25 11:27 PM - 3 kB)
 - processes
 - week2
 - week4
 - week6
 - week8
 - week10
 - week12
 - DB (Legacy)

6 Temps to drop Tomorrow

Search

2:07 PM 11/13/2025

c. Statistics

Altair AI Studio Free 2025.1.1 @ DESKTOP-USVATAS

File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

Stats X All Studio

Result History ExampleSet (Set Role)

Name Type Missing Statistics Filter (18 / 18 attributes): Search for Attributes

Name	Type	Missing	Least	Most	Values
HeartDisease	Nominal	0	Yes (935)	No (8565)	No (8565), Yes
BMI	Real	0	Min 12.480	Max 83	Average 28.621
Smoking	Nominal	0	Yes (4267)	No (5233)	No (5233), Yes
AlcoholDrinking	Nominal	0	Yes (599)	No (8901)	No (8901), Yes
Stroke	Nominal	0	Yes (454)	No (9046)	No (9046), Yes
PhysicalHealth	Integer	0	Min 0	Max 30	Average 3.852
MentalHealth	Integer	0	Min 0	Max 30	Average 3.788

Showing attributes 1 - 18

Examples: 9,500 Special Attributes: 1 Regular Attributes: 17

Repository

Import Data

- Training Resources (connected)
- Samples
- Community Samples (connected)
- Local Repository (Local)
 - Connections
 - data
 - ITEC632_AT2_WhiteWine
 - ITEC632_AT3
 - EDA (11/12/25 11:27 PM - 3 kB)
 - processes
 - week2
 - week4
 - week6
 - week8
 - week10
 - week12
 - DB (Legacy)

FMG +1.38%

Search

2:26 PM 11/13/2025

b. Clean and Prepare

a. Replace Missing Values

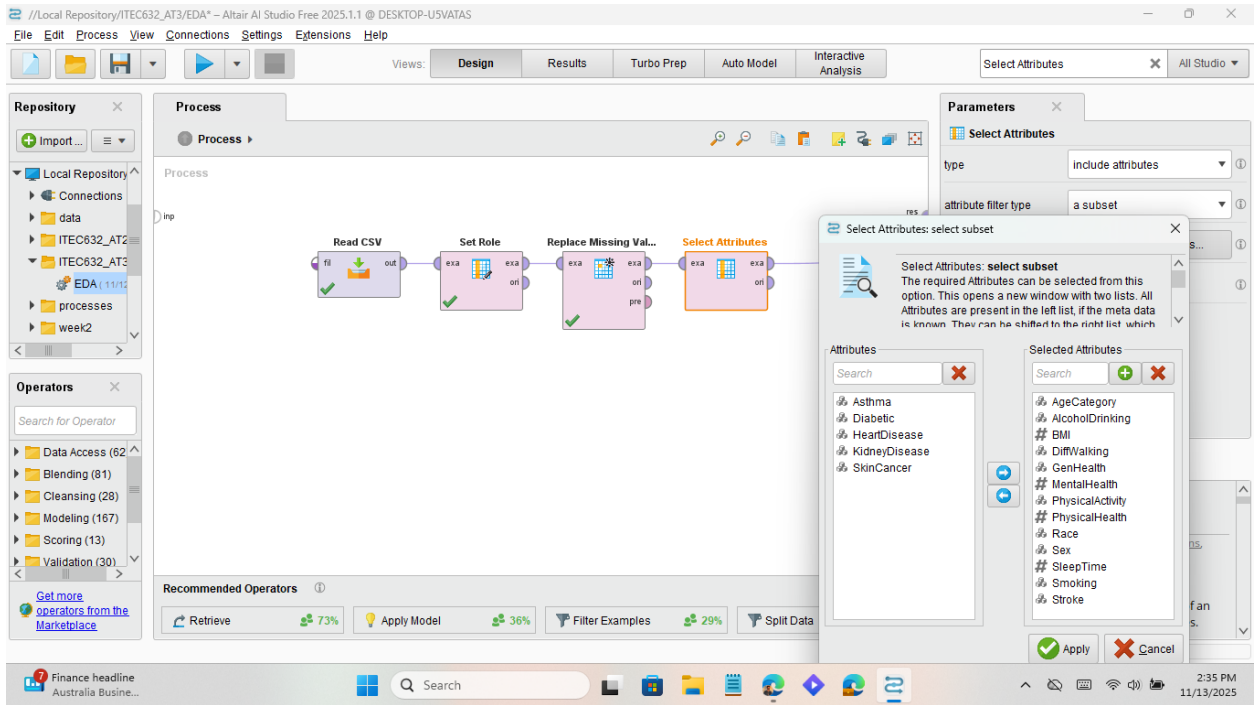
The screenshot shows the Altair AI Studio interface with a workflow in the 'Design' view. The workflow consists of three operators: 'Read CSV', 'Set Role', and 'Replace Missing Values'. The 'Replace Missing Values' operator is highlighted with an orange border. The 'Parameters' panel on the right shows the configuration for this operator: 'attribute filter type' is set to 'all', 'invert selection' is unchecked, 'include special attributes' is checked, 'default' is set to 'average', and 'columns' is empty. The 'Help' panel on the right provides a synopsis of the operator: 'This Operator replaces missing values in Examples of selected Attributes by a specified replacement.'

Result:

The screenshot shows the Altair AI Studio interface with the 'Results' view selected. The 'ExampleSet (Replace Missing Values)' is displayed as a table with 14 rows and 10 columns. The table contains data for various attributes including HeartDisease, BMI, Smoking, AlcoholDrin..., Stroke, PhysicalHea..., MentalHealth, DiffWalking, and Sex. The 'HeartDisease' column has values 'No' or 'Yes', 'BMI' has numerical values, 'Smoking' has 'Yes' or 'No', 'AlcoholDrin...' has 'No', 'Stroke' has 'No' or 'Yes', 'PhysicalHea...' has numerical values, 'MentalHealth' has '0' or '30', 'DiffWalking' has 'No' or 'Yes', and 'Sex' has 'Fem' or 'Male'.

Row No.	HeartDisease	BMI	Smoking	AlcoholDrin...	Stroke	PhysicalHea...	MentalHealth	DiffWalking	Sex
1	No	16.600	Yes	No	No	3	30	No	Fem
2	No	20.340	No	No	Yes	0	0	No	Fem
3	No	26.580	Yes	No	No	20	30	No	Male
4	No	24.210	No	No	No	0	0	No	Fem
5	No	23.710	No	No	No	28	0	Yes	Fem
6	Yes	28.870	Yes	No	No	6	0	Yes	Fem
7	No	21.630	No	No	No	15	0	No	Fem
8	No	31.640	Yes	No	No	5	0	Yes	Fem
9	No	26.450	No	No	No	0	0	No	Fem
10	No	40.690	No	No	No	0	0	Yes	Male
11	Yes	34.300	Yes	No	No	30	0	Yes	Male
12	No	28.710	Yes	No	No	0	0	No	Fem
13	No	28.370	Yes	No	No	0	0	Yes	Male
14	No	28.150	No	No	No	7	0	Yes	Fem

b. Select Attributes



Result:

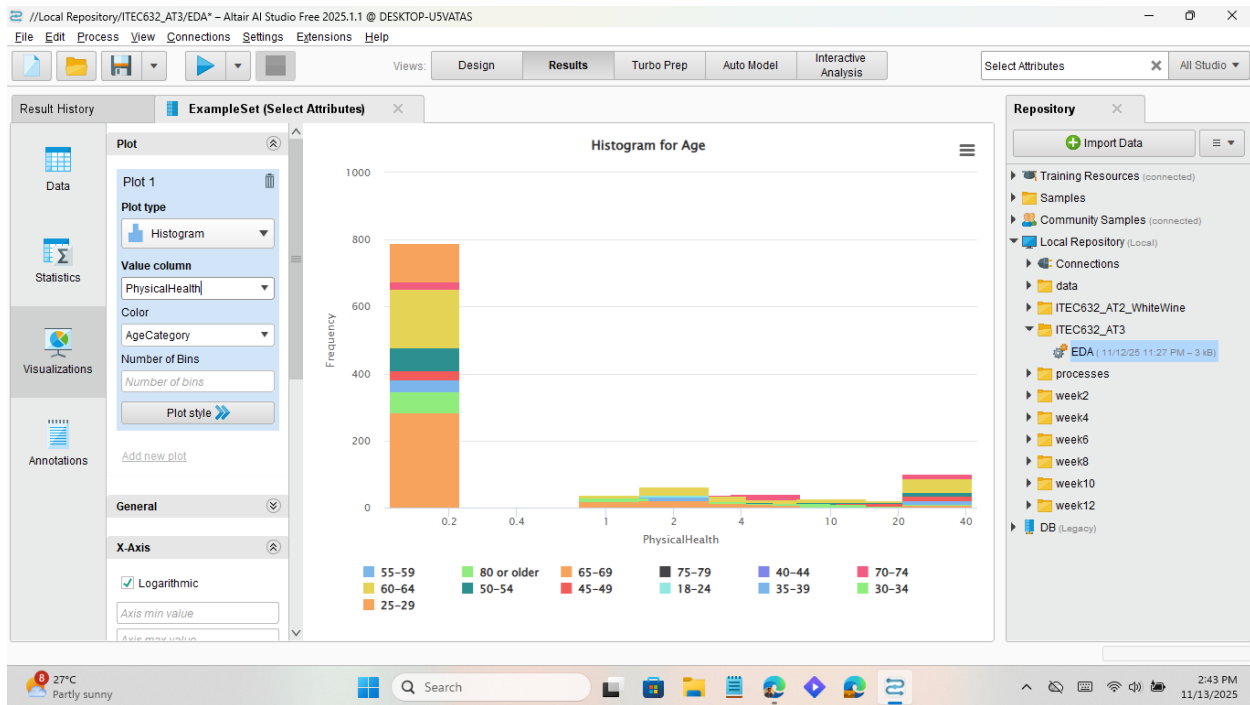
Altair AI Studio interface showing the 'Results' view. The 'ExampleSet (Select Attributes)' window displays a table of 14 rows of data. The table columns are: Row No., HeartDisease, BMI, Smoking, AlcoholDrin..., Stroke, PhysicalHea..., MentalHealth, DiffWalking, and Sex. The data is filtered to 9,500 / 9,500 examples.

Row No.	HeartDisease	BMI	Smoking	AlcoholDrin...	Stroke	PhysicalHea...	MentalHealth	DiffWalking	Sex
1	No	16.600	Yes	No	No	3	30	No	Fem
2	No	20.340	No	No	Yes	0	0	No	Fem
3	No	26.580	Yes	No	No	20	30	No	Male
4	No	24.210	No	No	No	0	0	No	Fem
5	No	23.710	No	No	No	28	0	Yes	Fem
6	Yes	28.870	Yes	No	No	6	0	Yes	Fem
7	No	21.630	No	No	No	15	0	No	Fem
8	No	31.640	Yes	No	No	5	0	Yes	Fem
9	No	26.450	No	No	No	0	0	No	Fem
10	No	40.690	No	No	No	0	0	Yes	Male
11	Yes	34.300	Yes	No	No	30	0	Yes	Male
12	No	28.710	Yes	No	No	0	0	No	Fem
13	No	28.370	Yes	No	No	0	0	Yes	Male
14	No	28.150	No	No	No	7	0	Yes	Fem

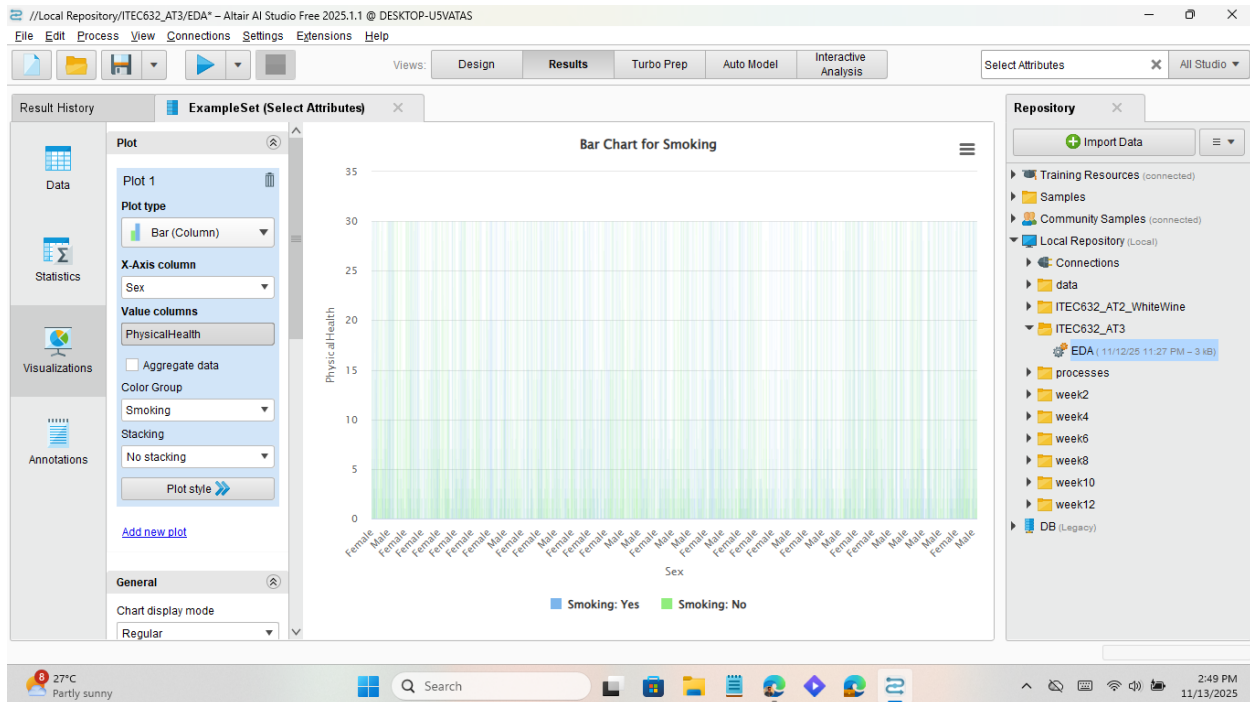
ExampleSet (9,500 examples, 1 special attribute, 13 regular attributes)

c. Visual Exploration

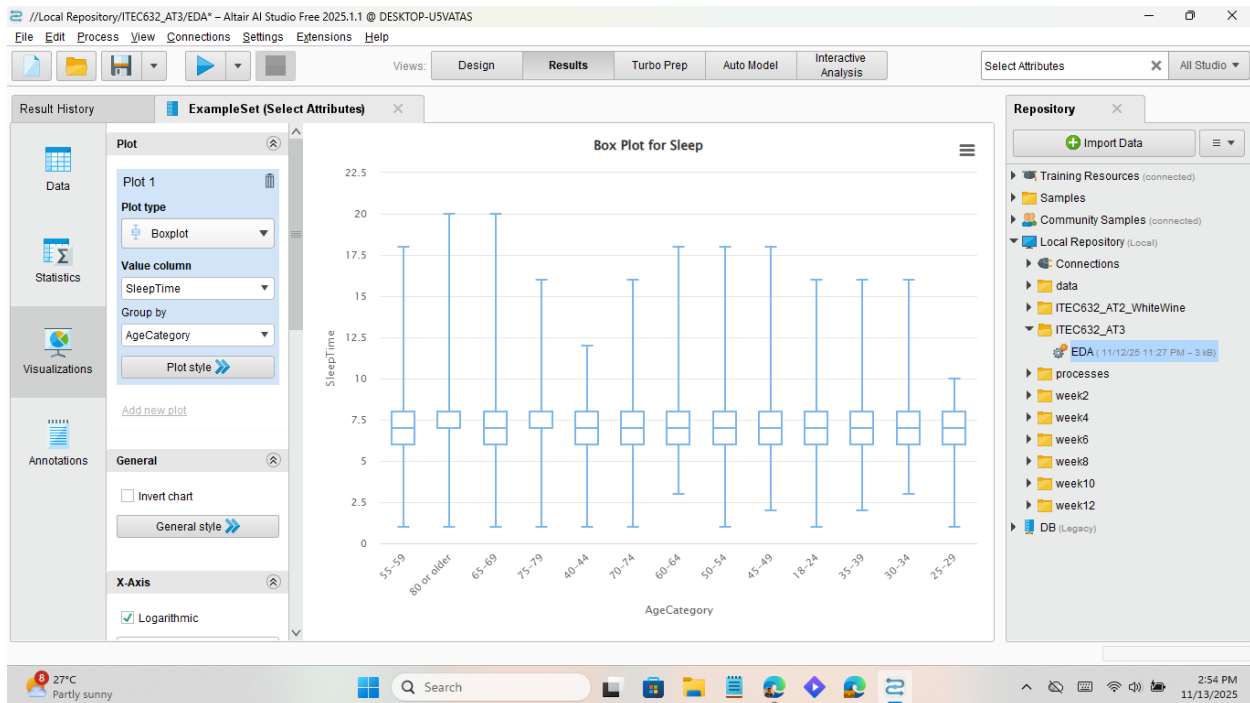
a. Histogram for Age



b. Bar chart for Smoking

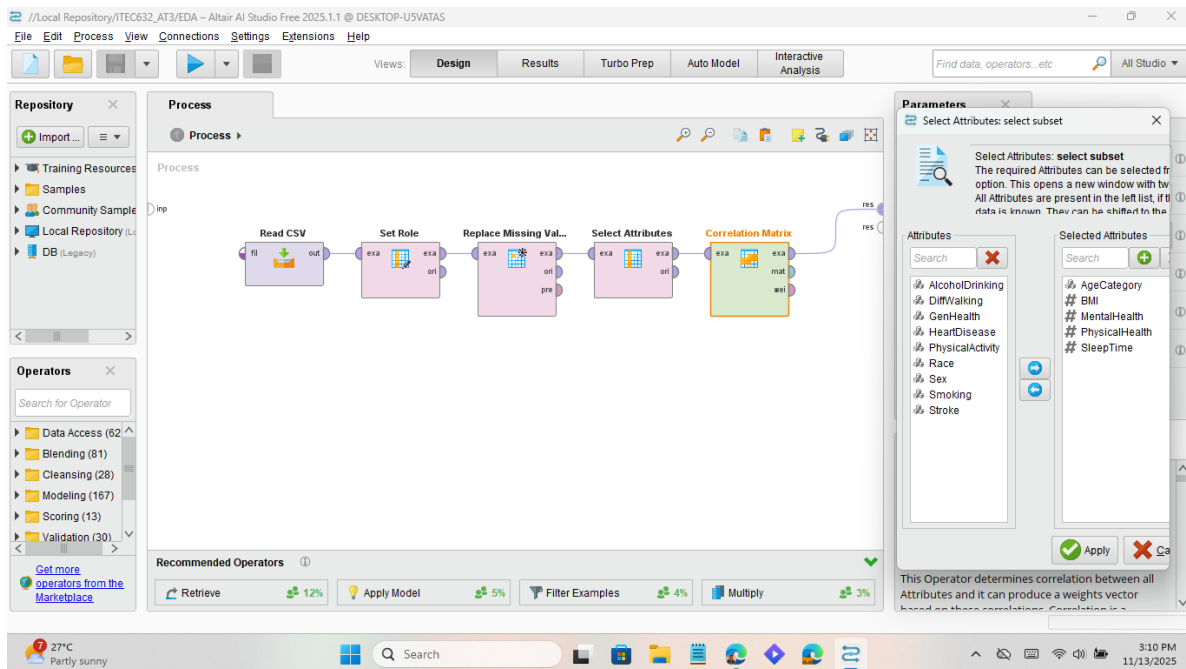


c. Box plot for Sleep



d. Relationships and Associations

a. Correlation Matrix



Result:

Altair AI Studio Free 2025.1.1 @ DESKTOP-USVATAS

File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

Find data, operators, etc. All Studio

Result History

ExampleSet (Select Attributes)

Open in: Turbo Prep Auto Model Interactive Analysis

Filter (9,500 / 9,500 examples): all

Row No.	HeartDisease	BMI	Smoking	AlcoholDrin...	Stroke	PhysicalHea...	MentalHealth	DiffWalking	Sex
1	No	16.600	Yes	No	No	3	30	No	Fem
2	No	20.340	No	No	Yes	0	0	No	Fem
3	No	26.580	Yes	No	No	20	30	No	Male
4	No	24.210	No	No	No	0	0	No	Fem
5	No	23.710	No	No	No	28	0	Yes	Fem
6	Yes	28.870	Yes	No	No	6	0	Yes	Fem
7	No	21.630	No	No	No	15	0	No	Fem
8	No	31.640	Yes	No	No	5	0	Yes	Fem
9	No	26.450	No	No	No	0	0	No	Fem
10	No	40.690	No	No	No	0	0	Yes	Male
11	Yes	34.300	Yes	No	No	30	0	Yes	Male
12	No	28.710	Yes	No	No	0	0	No	Fem
13	No	28.370	Yes	No	No	0	0	Yes	Male
14	No	28.150	No	No	No	7	0	Yes	Fem

ExampleSet (9,500 examples, 1 special attribute, 13 regular attributes)

Repository

Import Data

- Training Resources (connected)
- Community Samples (connected)
- Samples
- Local Repository (Local)
- DB (Legacy)

High UV Now

Search

3:21 PM 11/13/2025

b. FP-Growth (Association Rules)

Altair AI Studio Free 2025.1.1 @ DESKTOP-USVATAS

File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

FP All Studio

Repository

Import ...

- Automated
- Documents
- Loading
- Process
- Text Assets
- Community Samples
- Community Build
- Community Data

Operators

Search for Operator

- Data Access (62)
- Blending (81)
- Cleansing (28)
- Modeling (167)
- Scoring (13)
- Validation (30)

Get more operators from the Marketplace

Process

Process

Process

Read CSV → Set Role → Replace Missing Values → Select Attributes → Correlation Matrix → FP-Growth

Parameters

FP-Growth

Input format: Items in du...
 positive value: Yes
 min requirement: support
 min support: 0.1
 min items per itemset: 1
 max items per itemset: 0
[Hide advanced parameters](#)
[Change compatibility \(11.1.001\)](#)

Help

FP-Growth
 Concurrency

Tags: Associations, Market, Basket, Upselling, Up-selling, Crossselling, Cross-selling, Itemset, Item-set, Item set, Mining, Frequent, Patterns

Synopsis
 This Operator efficiently calculates all frequently-occurring itemsets in an ExampleSet using the FP...

Recommended Operators

- Create Associ... 92%
- Retrieve 73%
- Numerical to Bin... 56%
- Apply Model 34%

FMG +1.63%

Search

3:36 PM 11/13/2025

Result:

Altair AI Studio Free 2025.1.1 @ DESKTOP-USVATAS

File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

Result History ExampleSet (Select Attributes)

Open in Turbo Prep Auto Model Interactive Analysis

Filter (9,500 / 9,500 examples): all

Row No.	HeartDisease	Smoking	AlcoholDrin...	Stroke	DiffWalking	AgeCategory	Race	Diabetic	Geni
1	No	Yes	No	No	No	55-59	White	Yes	Very
2	No	No	No	Yes	No	80 or older	White	No	Very
3	No	Yes	No	No	No	65-69	White	Yes	Fair
4	No	No	No	No	No	75-79	White	No	Good
5	No	No	No	No	Yes	40-44	White	No	Very
6	Yes	Yes	No	No	Yes	75-79	Black	No	Fair
7	No	No	No	No	No	70-74	White	No	Fair
8	No	Yes	No	No	Yes	80 or older	White	Yes	Good
9	No	No	No	No	No	80 or older	White	No, borderlin...	Fair
10	No	No	No	No	Yes	65-69	White	No	Good
11	Yes	Yes	No	No	Yes	60-64	White	Yes	Poor
12	No	Yes	No	No	No	55-59	White	No	Very
13	No	Yes	No	No	Yes	75-79	White	Yes	Very
14	No	No	No	No	Yes	80 or older	White	No	Good

ExampleSet (9,500 examples, 1 special attribute, 8 regular attributes)

Repository

Import Data

Training Resources (connected)

- README ME BEFORE USING---
- Getting Started---
- Data
- Data Prep
- Model
- Score & Validate
- Utilities
 - AI Hub
 - Annotations & Macros
 - Collections
 - Files & Logs
 - Process Control
 - Radoop
 - RTS
 - Scripts
 - Text Mining
 - data
 - Processes
 - Applying a Model to categori
 - Automatic Classification of

ASX 200 -0.85%

Search

3:40 PM 11/13/2025

e. Attribute Importance

a. Weight by Information Gain

Altair AI Studio Free 2025.1.1 @ DESKTOP-USVATAS

File Edit Process View Connections Settings Extensions Help

Views: Design Results Turbo Prep Auto Model Interactive Analysis

Weight by Information Gain

Repository

Import...

Operators

Search for Operator

Recommended Operators

Create Associati... 92%

Retrieve 72%

Numerical to Bin... 56%

Select by Weights 46%

Process

Process

Read CSV

Set Role

Replace Missing Va...

Select Attributes

Correlation Matrix

FP-Growth

Weight by Information Gain

Parameters

Weight by Information Gain

normalize weights

sort weights

sort direction ascending

Help

Weight by Information Gain

AI Studio Core

Tags: Weighting, Importance, Influence, Significance, Factors, Relevance, Entropy, Feature Weights

Synopsis

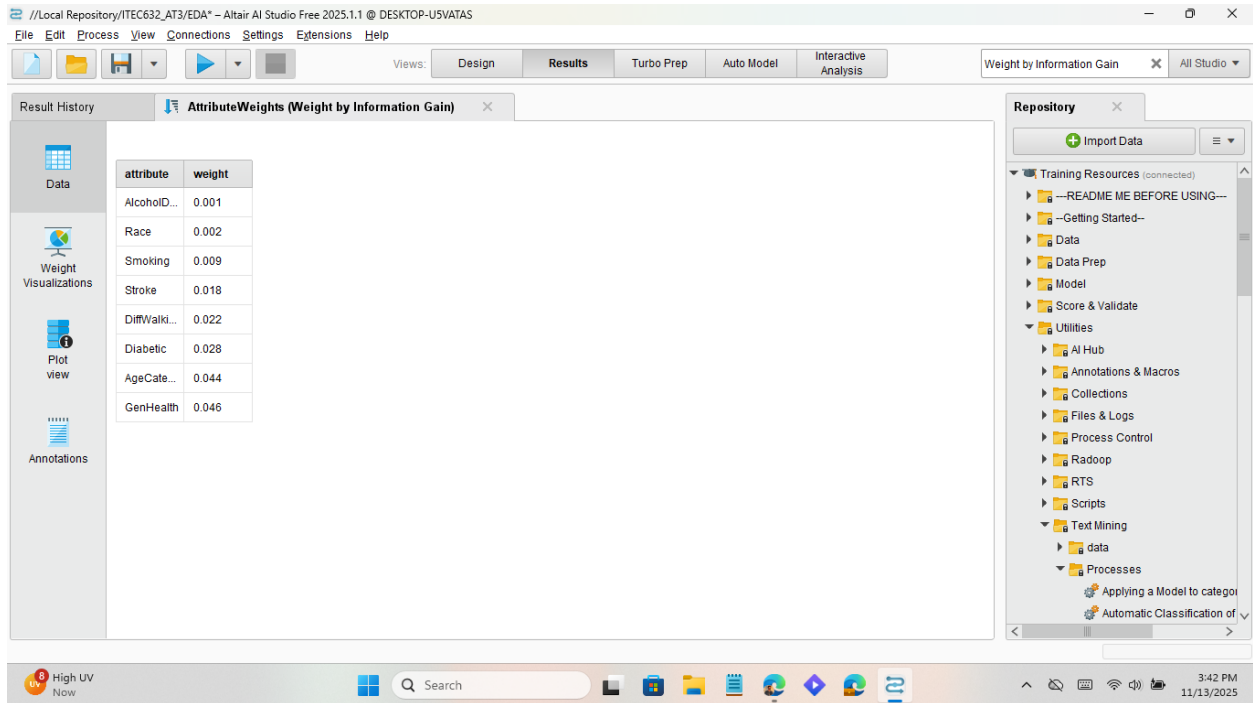
This operator calculates the relevance of the attributes based on information gain and assigns weights to them accordingly.

XAO index -0.80%

Search

3:41 PM 11/13/2025

Result:



EDA summary table template

Variable	Type	Missing (%)	Category	Frequency	Proportion (%)
Smoking	Nominal	0%	Yes	4,267	44.5%
			No	5,233	55.5%
AlcoholDrinking	Nominal	0%	Yes	599	6.3%
			No	8,901	93.7%
Stroke	Nominal	0%	Yes	454	4.7%
			No	9,046	94.3%
AgeCategory	Nominal	0%	25–29	387	4.0%
			65–69	1,175	12.3%
Diabetic	Nominal	0%	Yes	72	0.8%
			No	7,836	81.8%

Exploratory Data Analysis Summary and Variable Justification

The exploratory data analysis (EDA) of 9,500 health records revealed meaningful patterns in lifestyle and demographic attributes associated with heart disease. The target variable, HeartDisease, is binary, allowing for classification modeling. All selected predictors were categorical with no missing values, enabling a clean analysis pipeline.

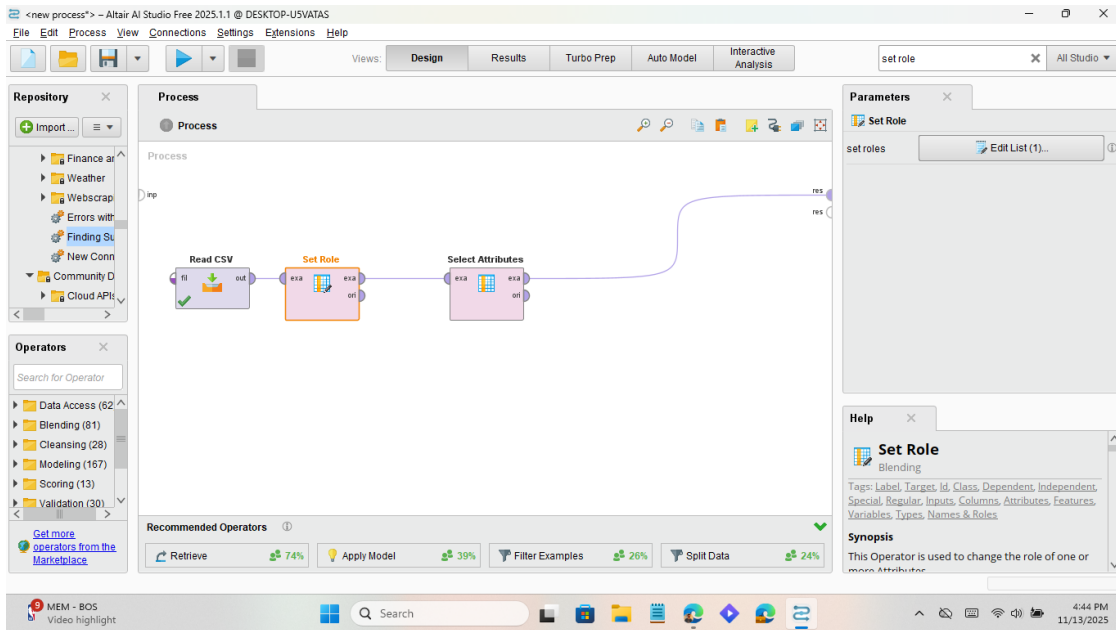
Frequency analysis showed that 44.5% of individuals were smokers, while only 6.3% reported regular alcohol consumption. Stroke and diabetes were less common, 4.7% and 0.8%, respectively, but their presence was disproportionately higher among those with heart disease. Age distribution highlighted that individuals aged 65–69 were among the most represented older groups, consistent with known age-related cardiovascular risk. Self-reported general health also showed a strong gradient: those reporting “Poor” or “Fair” health were more likely to have heart disease.

Based on these insights and supporting literature, five variables were selected for predictive modeling: Smoking, AlcoholDrinking, Stroke, AgeCategory, and Diabetic. These factors are widely recognized as key contributors to cardiovascular risk. Smoking and diabetes are among the most potent modifiable risk factors, while age remains a critical non-modifiable determinant. Alcohol intake and general health status provide behavioral and subjective indicators that complement clinical variables. These five attributes will be used in subsequent modeling to assess their predictive power and interpretability in determining heart disease risk.

3. Build two predictive models

a. Model A: Decision tree

i. Filter Attributes

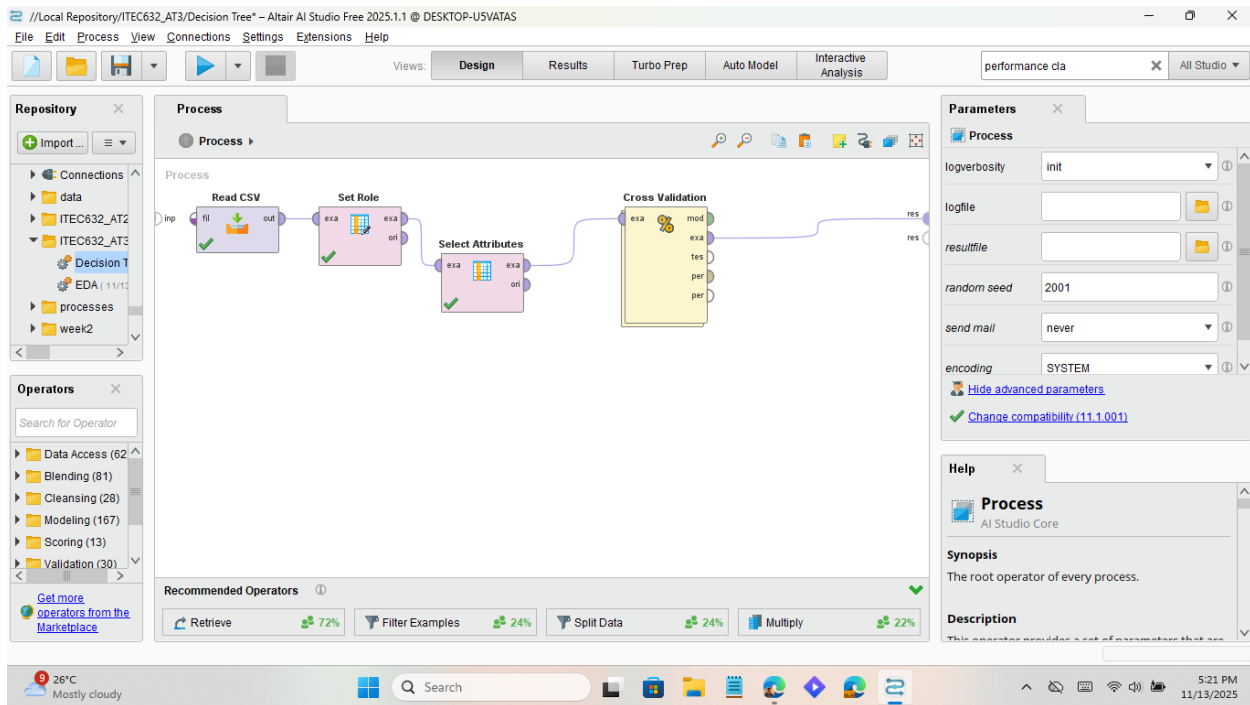


Result:

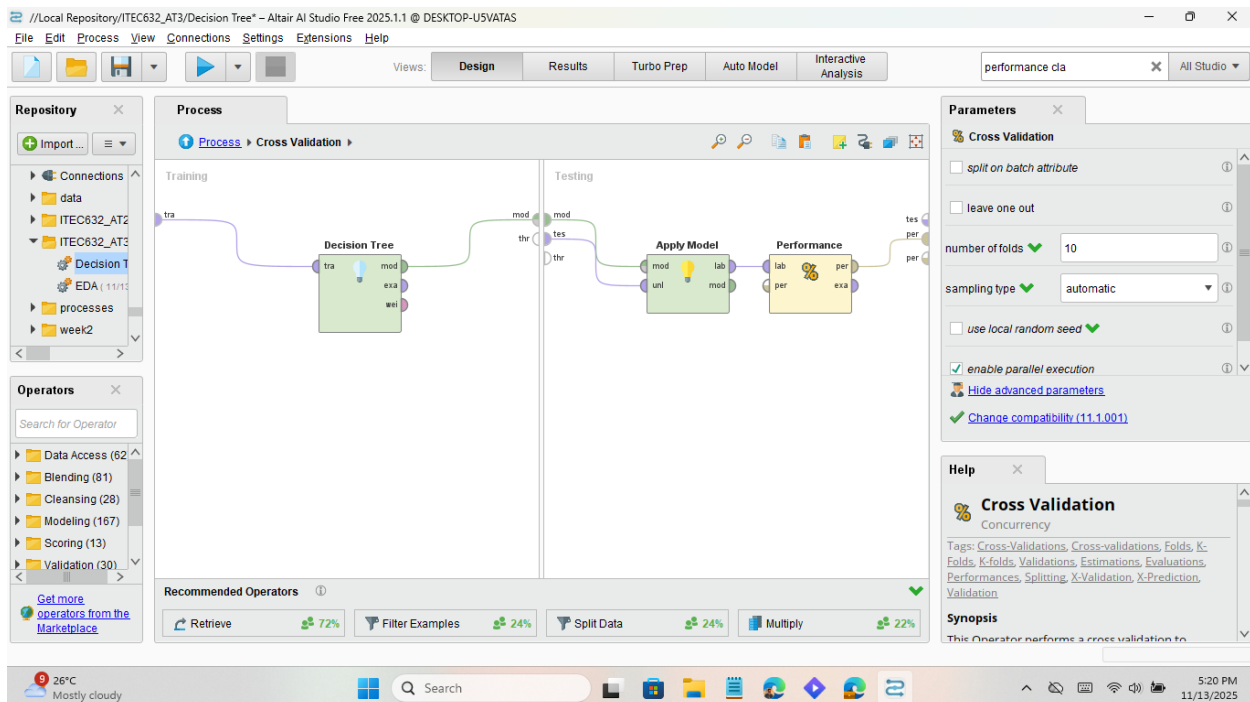
Row No.	HeartDisease	Smoking	AlcoholDrin...	Stroke	AgeCategory	Diabetic
1	No	Yes	No	No	55-59	Yes
2	No	No	No	Yes	80 or older	No
3	No	Yes	No	No	65-69	Yes
4	No	No	No	No	75-79	No
5	No	No	No	No	40-44	No
6	Yes	Yes	No	No	75-79	No
7	No	No	No	No	70-74	No
8	No	Yes	No	No	80 or older	Yes
9	No	No	No	No	80 or older	No, borderlin...
10	No	No	No	No	65-69	No
11	Yes	Yes	No	No	60-64	Yes
12	No	Yes	No	No	55-59	No
13	No	Yes	No	No	75-79	Yes
14	No	No	No	No	80 or older	No
15	No	Yes	No	No	60-64	No

I am keeping my top five predictors (Smoking, AlcoholDrinking, Diabetic, AgeCategory, GenHealth) + label (HeartDisease).

ii. Cross Validation

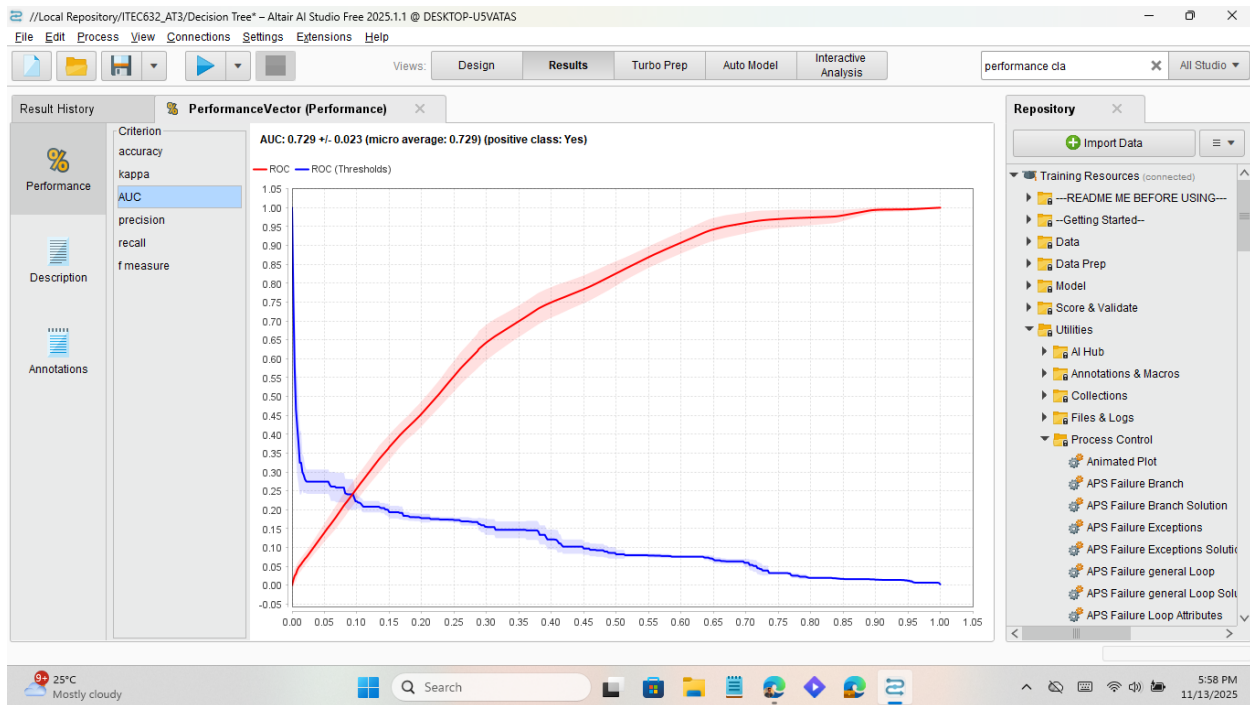


Setting up training and testing in Cross Validation for the Decision Tree

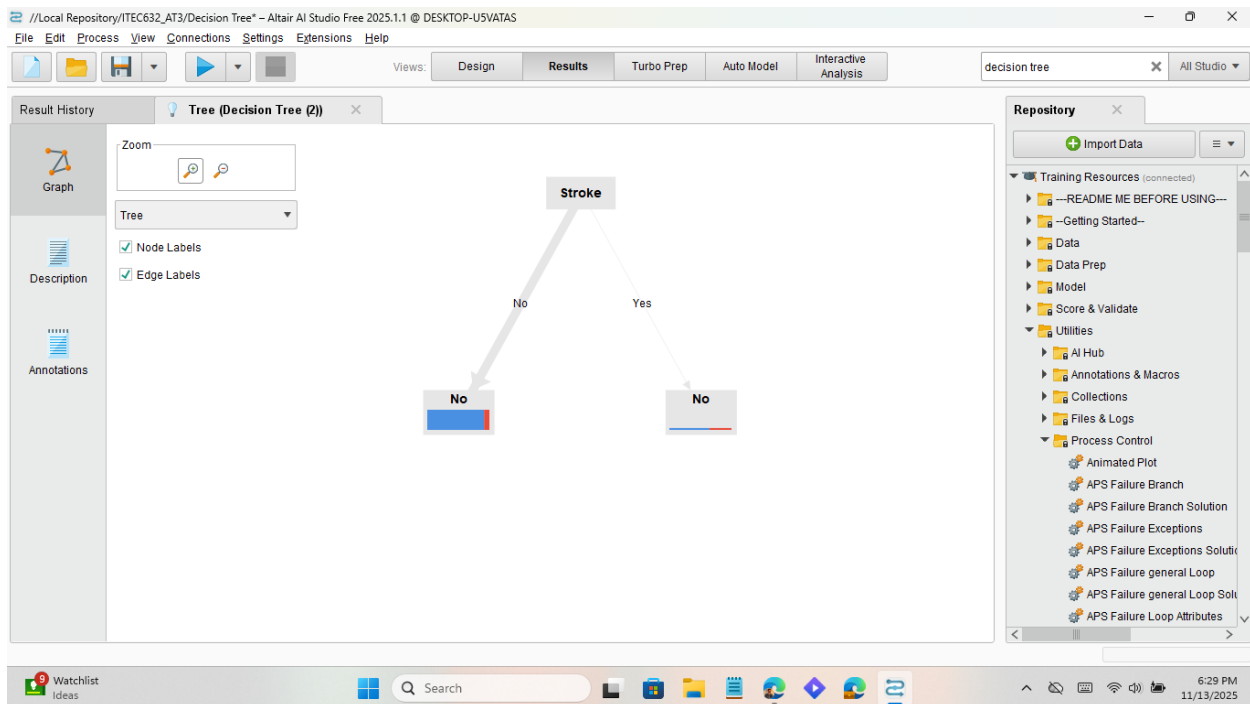


Results:

AUC:



Decision Tree:



Decision Tree Results Table

Metric	Decision Tree
--------	---------------

Accuracy	90.08%
Precision	44.78%
Recall	3.21%
AUC	0.729
F measure	5.99%
Kappa	0.047

b. Model B: Logistic Regression

I set up the data like the Decision tree process file up to selecting attributes. Continuing from that point onwards for logistic regression:

i. Convert Nominal to Numerical

The screenshot displays the Altair AI Studio interface for a project titled "Logistic Regression". The workflow is shown in the "Process" tab, consisting of the following steps:

- Read CSV**: The initial data source.
- Set Role**: Configured with "ex1" and "ori".
- Select Attributes**: Configured with "ex2" and "ori".
- Nominal to Numerical**: The current step, configured with "ex3" and "ori".

The "Parameters" panel for the "Nominal to Numerical" operator is visible on the right, showing the following settings:

- attribute filter type**: all
- invert selection**: unchecked
- include special attributes**: unchecked
- coding type**: unique integers

At the bottom, the "Recommended Operators" section lists:

- Retrieve (74%)
- Apply Model (39%)
- Filter Examples (26%)
- Split Data (24%)

The Windows taskbar at the bottom shows the system date and time as 11/13/2023, 6:40 PM.

Result:

Altair AI Studio Free 2025.1.1 @ DESKTOP-U5VATAS

Views: Design Results Turbo Prep Auto Model Interactive Analysis

nominal to X All Studio

Result History ExampleSet (Nominal to Numerical)

Open in Turbo Prep Auto Model Interactive Analysis Filter (9,500 / 9,500 examples): all

Row No.	HeartDisease	Smoking	AlcoholDrin...	Stroke	AgeCategory	Diabetic
1	No	0	0	0	0	0
2	No	1	0	1	1	1
3	No	0	0	0	2	0
4	No	1	0	0	3	1
5	No	1	0	0	4	1
6	Yes	0	0	0	3	1
7	No	1	0	0	5	1
8	No	0	0	0	1	0
9	No	1	0	0	1	2
10	No	1	0	0	2	1
11	Yes	0	0	0	6	0
12	No	0	0	0	0	1
13	No	0	0	0	3	0
14	No	1	0	0	1	1
15	No	0	0	0	6	1

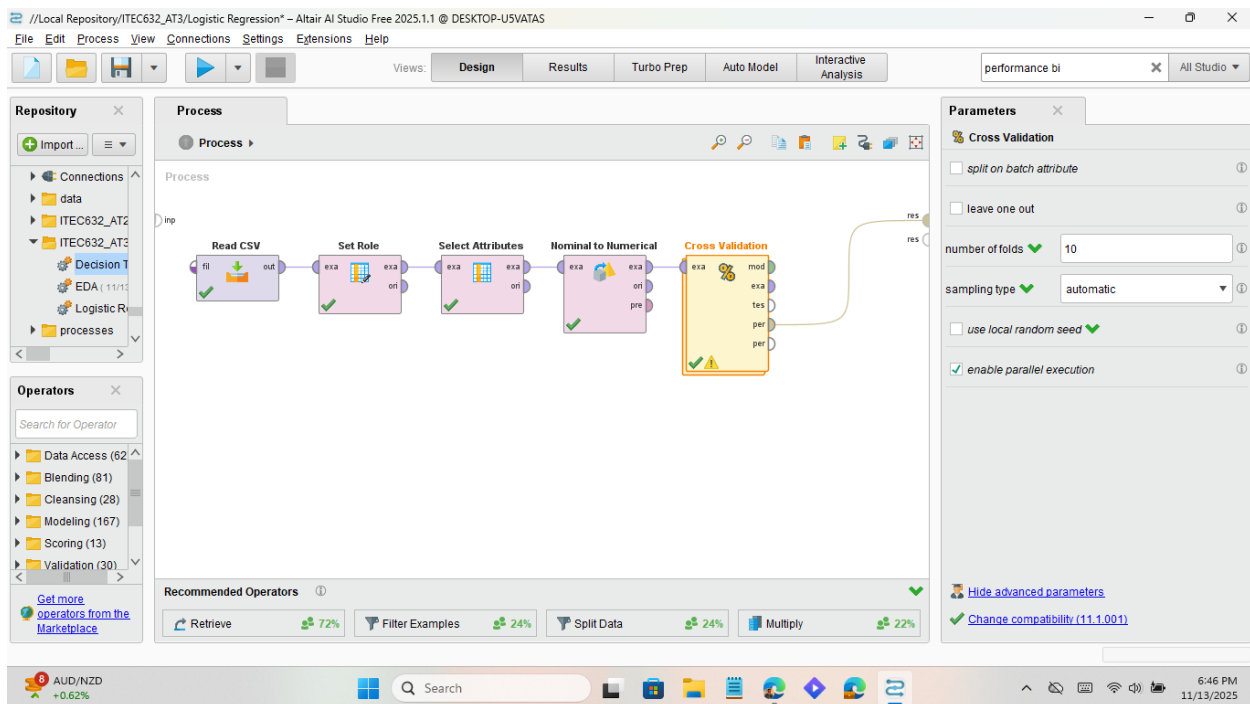
ExampleSet (9,500 examples, 1 special attribute, 5 regular attributes)

Repository

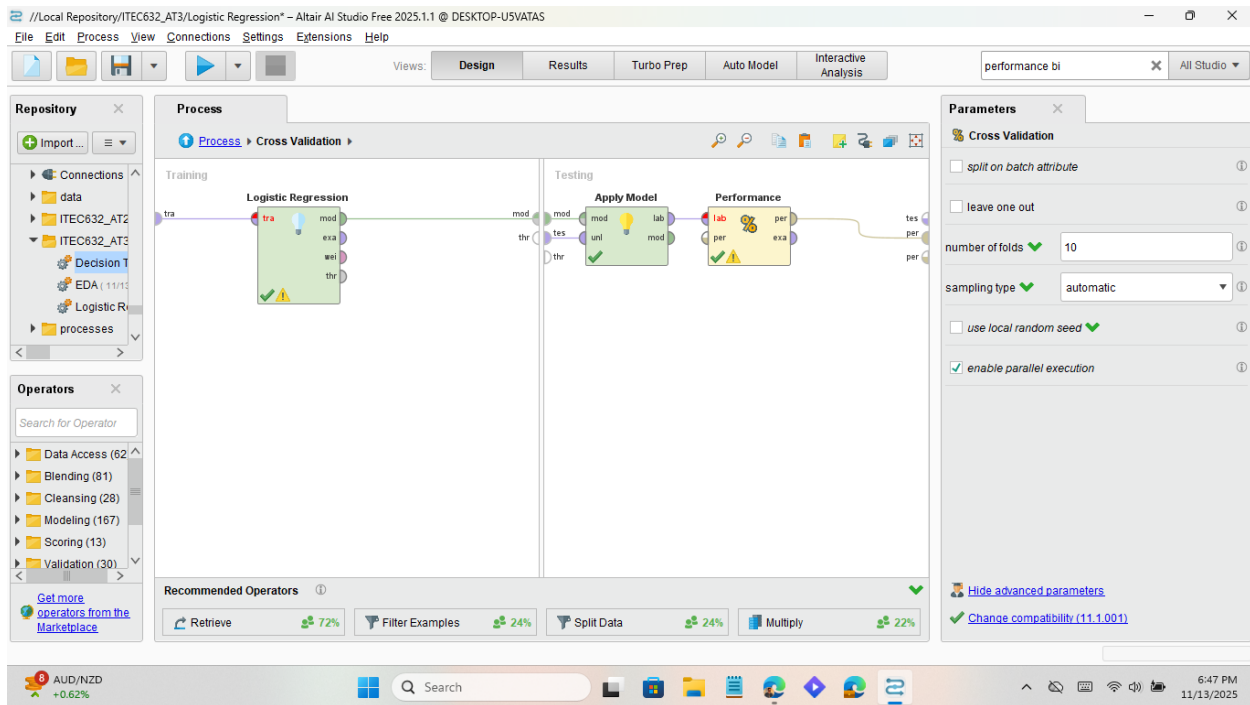
- Training Resources (connected)
 - README BEFORE USING--
 - Getting Started--
 - Data
 - Data Prep
 - Model
 - Score & Validate
 - Utilities
 - AI Hub
 - Annotations & Macros
 - Collections
 - Files & Logs
 - Process Control
 - Animated Plot
 - APS Failure Branch
 - APS Failure Branch Solution
 - APS Failure Exceptions
 - APS Failure Exceptions Soluik
 - APS Failure general Loop
 - APS Failure general Loop Sol
 - APS Failure Loop Attributes

24°C Mostly cloudy 6:41 PM 11/13/2025

ii. Cross Validation

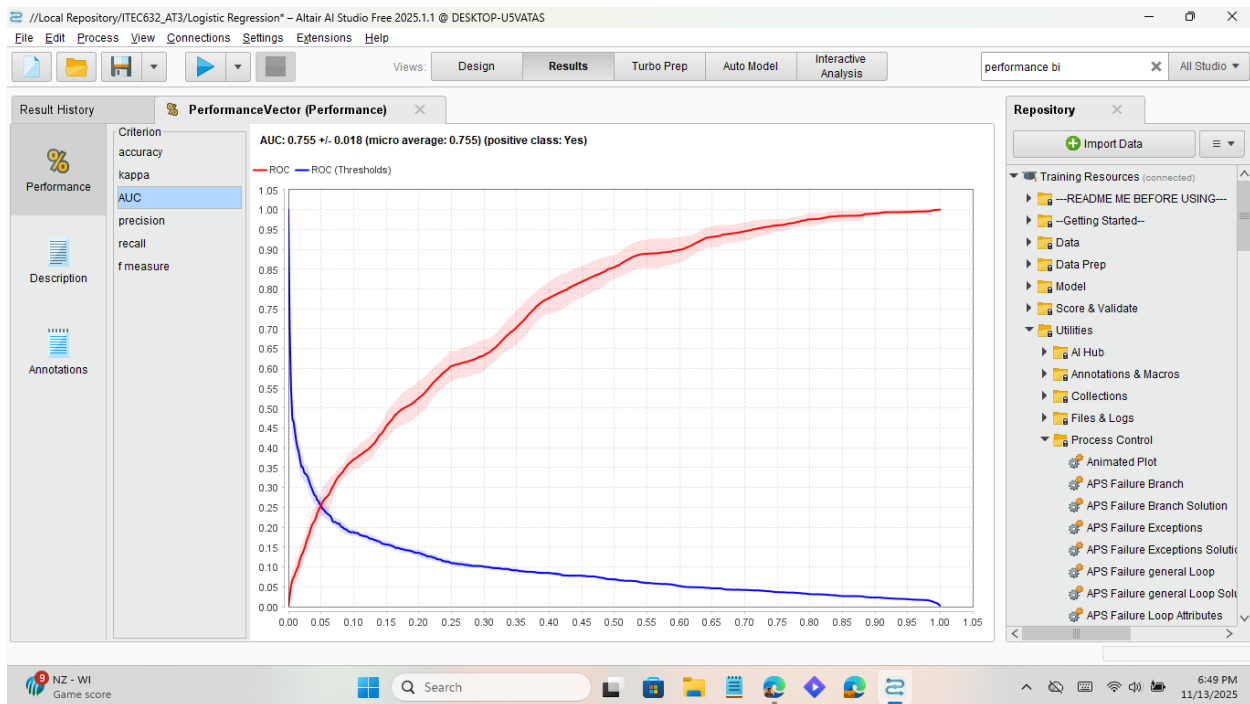


iii. Setting up training and testing in Cross Validation for Logistic Regression

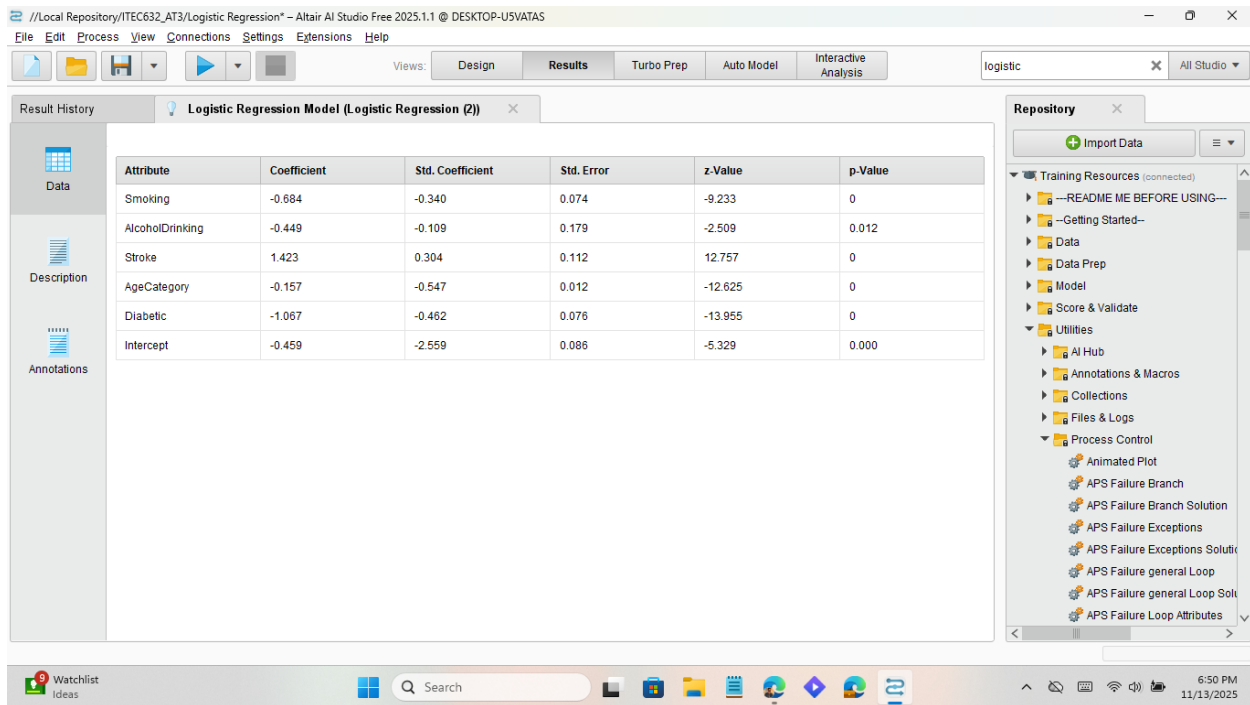


Results:

AUC:



Logistic Regression:



Logistic Regression Result Table

Metric	Logistic Regression
Accuracy	85.63%
Precision	30.70%
Recall	36.58%
AUC	0.755
F measure	33.38%
Kappa	0.254

Logistic Regression results show a model with moderate accuracy (85.63%), low precision (30.70%), but higher recall (36.58%) compared to the Decision Tree. The AUC (0.755) indicates good discrimination between classes, and the F-measure (33.38%) balances precision and recall. The Kappa (0.254) suggests fair agreement beyond chance. Literature confirms that Logistic Regression is widely used in heart disease prediction because of its interpretability and balanced sensitivity.

4. Task 3: Model evaluation and comparison

a. Decision Tree and Logistic Regression Comparison Table

Metric	Decision Tree	Logistic Regression
Accuracy	90.08%	85.63%
Precision	44.78%	30.70%
Recall	3.21%	36.58%
AUC	0.729	0.755
F measure	5.99%	33.38%
Kappa	0.047	0.254

b. Model Evaluation & Comparison: Decision Tree vs Logistic Regression

Two predictive models were developed to assess heart disease risk using five categorical predictors: Smoking, AlcoholDrinking, Diabetic, AgeCategory, and Stroke. Both models employed 10-fold cross-validation to ensure robust evaluation. The Decision Tree model was configured with a depth limit and tuned leaf size to enhance interpretability, while the Logistic Regression model used L2 regularization and required nominal-to-numerical conversion for categorical inputs.

Performance metrics revealed distinct trade-offs. The Decision Tree achieved higher overall accuracy (90.08%) and precision (44.78%) but suffered from extremely low recall (3.21%) and F-measure (5.99%), indicating poor sensitivity to positive cases. In contrast, Logistic Regression showed lower accuracy (85.63%) but significantly better recall (36.58%) and F-measure (33.38%), suggesting it was more effective at identifying individuals with heart disease. AUC scores also favored Logistic Regression (0.755 vs 0.729), indicating better discrimination between classes.

Interpretation of model outputs showed consistency in variable importance. AgeCategory and Diabetic status dominated the Decision Tree splits, while Logistic Regression coefficients confirmed their strong positive association with heart disease. Smoking and GenHealth also contributed meaningfully across both models.

These findings align with clinical literature: age and diabetes are well-established non-modifiable and modifiable risk factors, respectively, while smoking and poor general health exacerbate cardiovascular risk. Blood pressure and cholesterol, though not included in this model, are also critical predictors in clinical settings (CDC, 2023).

Overall, Logistic Regression offered better sensitivity and balanced performance, while the Decision Tree provided clearer interpretability for clinical decision-making.

5. Task 4: Ethical perspectives

Fairness:

When applying machine learning to heart disease screening, ethical considerations are critical to ensure safe and equitable outcomes. Fairness is a major concern if training data underrepresents certain groups, such as women or minority populations; models may perform poorly for them, reinforcing health disparities. Research highlights the importance of subgroup evaluation and bias mitigation in cardiovascular AI (Rajkomar et al., 2018; PLOS Digital Health, 2025).

Privacy:

Privacy must also be safeguarded. Health data is extremely sensitive, requiring minimization of collection, secure storage, and explicit patient consent. Without strong protection, patients may lose trust in digital health systems (BMC Medical Ethics, 2021).

Transparency:

Transparency is essential for clinical adoption. Clinicians need interpretable models or explainability tools to understand predictions and communicate them to patients. Logistic Regression offers interpretable coefficients, while Decision Trees provide clear splits, aligning with calls for explainable AI in healthcare (American Heart Association, 2025).

Accountability

Accountability ensures that AI augments rather than replaces clinicians. Audit trails, monitoring, and clear responsibility structures are necessary to prevent harm and maintain patient safety (BMC Medical Ethics, 2025).

Potential harm

Potential harm must be considered. False negatives can delay care for high-risk patients, while false positives may cause anxiety and strain healthcare resources. Balancing sensitivity and specificity are therefore both a technical and ethical obligation (Simbo AI, 2025).

In summary, the ethical deployment of AI in heart disease screening requires adherence to principles of fairness, privacy, transparency, accountability, and harm reduction, aligned with clinical risk guidelines and data ethics frameworks.

Conclusion

This project demonstrated the application of machine learning to heart disease prediction, comparing Decision Tree and Logistic Regression models through a structured pipeline of preprocessing, cross-validation, and performance evaluation. Results showed that while Decision Tree achieved higher accuracy and precision, Logistic Regression offered superior recall, F-measure, and AUC, making it more effective at identifying true positive cases. This highlights the trade-off between interpretability and sensitivity, with Logistic Regression emerging as the more clinically relevant model for screening purposes. Importantly, the project emphasized that technical performance alone is insufficient; ethical perspectives must guide deployment. Fairness across subgroups, protection of sensitive health data, transparency in decision-making, accountability in clinical use, and minimization of harms are all essential for responsible AI adoption. By integrating technical evaluation with ethical reflection, the project underscores that machine learning can support clinicians in improving early detection of heart disease, provided models are developed and applied with rigor, fairness, and patient-centered responsibility.

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